Visual-Prompt Bidirectional Interactive Learning for Remote Sensing Infrared and Visible Image Fusion

Fan Zhao, Yidi Wang, Wenda Zhao, *Member, IEEE*, and Jiao Li

***Abstract*—Remote sensing infrared and visible image fusion (IVIF) aims to leverage the modality advantages of infrared and visible images to enhance the comprehensiveness and precision of visual representations. However, existing text-guided methods are only constrained by the unidirectional text-to-vision semantic guidance paradigm, where textual description biases directly propagate to fusion results. This may lead to deviations in the fusion outputs, thereby suffering from the loss of original visual information. Therefore, this article proposes a visual-prompt bidirectional interactive learning (VP-BIL) method for remote sensing IVIF. The method constructs a bidirectional semantic alignment mechanism based on visual features and prompt representations generated by the text encoder, enabling efficient cross-modal fusion without requiring accurate text annotations. In particular, a text-visual semantic correlation is first established that ensures both semantic alignment accuracy and the preservation of original visual information. Subsequently, a visual-driven text correction module is designed to dynamically calibrate textual description biases, enhancing the cross-modal semantic alignment between the text and vision. Extensive experiments demonstrate that the proposed method effectively overcomes the unidirectional dependence limitation of the text-guided paradigm, outperforming the state-of-the-art (SOTA) methods.**

***Index Terms*—Bidirectional interactive learning, infrared and visible image fusion (IVIF), semantic alignment, visual-driven text correction.**

I. Introduction

REMOTE sensing infrared and visible image fusion (IVIF) aims to integrate the robust thermal target detection capability of infrared imagery and the abundant textural details of visible imagery, thereby yielding fused results with more comprehensive scene information. IVIF is crucial to downstream tasks, e.g., object detection [1], semantic segmentation [2], security surveillance [3], image segmentation [4], [5], [6], [7], and remote sensing change detection [8]. In remote sensing applications, the fusion results must simultaneously satisfy the dual requirements of accurate preservation of thermal targets under complex illumination conditions and clear representation of terrain texture details. This imposes stringent requirements on the semantic understanding capabilities of fusion methods [9].

In recent years, deep learning has driven the continuous advancement of IVIF methods beyond traditional paradigms, yet notable limitations persist for such methods, which can be categorized into two aspects. One category of methods primarily focuses on visual feature-driven approaches, which constrain the fusion process by designing complex encoder–decoder architectures [10], [11], transformer modules [12], [13], or handcrafted loss functions (e.g., structural similarity loss and gradient loss) [14], [15]. Although these approaches are effective in optimizing local textures and contrast, they neglect the deep semantic information embedded in the images. The fused outputs fail to align with the global semantic logic of remote sensing scenarios, such as the context-specific characteristics of mountain roads and farmland boundaries, thereby failing to meet the semantic consistency requirements essential to precise remote sensing tasks.

The other category centers on textual semantic guidance, which implements feature modulation via lightweight text encoding, or establishes text-guided fusion paradigms by leveraging the cross-modal comprehension capabilities of vision-language models (VLMs). For instance, Text-IF [16] leverages semantic text guidance for degradation-aware image fusion. The FILM framework [17] generates textual descriptions via ChatGPT. Then, the textual descriptions are used to guide the visual information fusion, which effectively enhances feature extraction and contextual understanding. Textfusion [18] and MTG Fusion [19] intend to incorporate high-level semantic information from the text modality to the vision level through visual-text interaction. However, such approaches adopt a unidirectional text-visual interaction paradigm that directly treats static textual information as semantic guidance, without fully considering the semantic inaccuracies of textual descriptions in practical remote sensing scenarios (e.g., omission of critical details such as “the intersection of waterways and roads” or “vegetation cover types”). The direct propagation of textual biases to the fused outputs, leading to the loss of thermal targets or texture features.

The existing text-guided fusion paradigms exhibit two typical limitations in their practical implementation. Methods based on Text-IF adopt a unidirectional pattern that employs textual features to modulate visual features, which easily leads to fusion detail degradation caused by textual bias propagation. Besides, VLM-based methods regard text as a fixed

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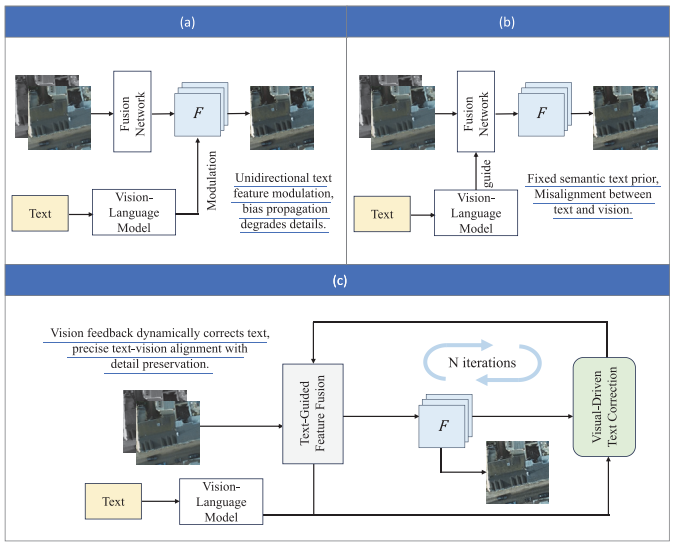


Fig. 1. Comparison of existing text-guided IVIF methods and the proposed VP-BIL method. (a) Text-IF methods. (b) VLM-based method. (c) Our method (VP-BIL).

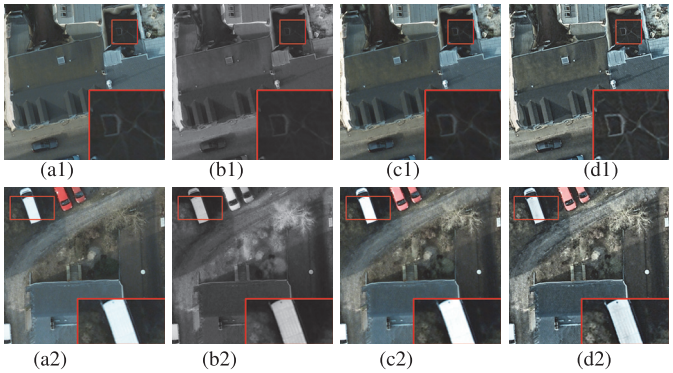


Fig. 2. Visual comparison between Text-IF and VP-BIL on Potsdam-TA. VP-BIL achieves semantic deviation calibration through bidirectional visual-prompt interaction, effectively mitigating detail degradation induced by textual biases compared with Text-IF. It ensures high-fidelity preservation of thermal targets and terrain textures, thereby exhibiting superior fusion performance. (a1) and (a2) Visible image. (b1) and (b2) Infrared image. (c1) and (c2) Text-IF [16]. (d1) and (d2) Ours.

semantic prior for fusion guidance, which hardly eliminates the inherent semantic misalignment between text and vision and thus results in inaccurate semantic alignment. Neither category establishes a bidirectional interaction mechanism between text and vision, making it difficult to fundamentally resolve fusion defects induced by inaccurate textual descriptions. To address the aforementioned issues, this article proposes a visual-prompt bidirectional interactive learning (VP-BIL) method. The core innovation lies in breaking the traditional unidirectional guidance paradigm and constructing an iterative closed-loop architecture with bidirectional linkage between text and vision. For an intuitive comparison of the structural differences between the proposed innovation and existing mainstream paradigms, the differentiated mechanism illustration is presented in Fig. 1. In particular, a text-guided feature fusion module (TGFF) is adopted, where the cross-attention mechanism [17] is introduced to achieve semantic anchoring. Subsequently, a visual-driven text correction module (VDTC) is designed to utilize deep fused visual features for reversely rectifying textual deviations. The two modules are integrated into a closed loop through multiple iterations. In each iteration, the derived fused features can calibrate textual representations and improve their representation accuracy. In turn, more accurate textual representations can further guide the fusion of original visual features to achieve superior fusion performance. The architecture enables precise cross-modal semantic matching without relying on elaborate text annotations.

The advantages of the proposed mechanism are intuitively reflected in visual comparison results, as shown in Fig. 2(c1) and (c2); the fusion results of Text-IF [16] exhibit obvious detail degradation. The texture of paved path intersections looks blurred and hazy, accompanied by severe attenuation of inherent intersecting structural features. Meanwhile, the striped texture on vehicle roofs becomes indistinct, resulting in massive loss of fine-grained details, which fails to accurately depict scene details and spatial features. On the contrary, the fusion results of VP-BIL exhibit clear and sharp structural layouts of paved path intersections, as well as distinct linear striped textures on vehicle roofs, as illustrated in Fig. 2(d1) and (d2). This method not only retains the fine textural details of visible images but also achieves precise alignment with the thermal radiation target characteristics of infrared images, thereby achieving a significant improvement in semantic compatibility.

Our contributions are summarized as follows.

1) To address the unidirectional dependence limitation of the text-guided paradigm, we propose a VP-BIL for IVIF.
2) A text-visual bidirectional interaction mechanism is designed for text-guided feature fusion through visual feedback-driven text correction, which preserves the textural fine details and achieves precise alignment with the thermal radiation target characteristics.
3) Extensive experiments are conducted on the remote sensing scene text–image paired datasets. The proposed VP-BIL method achieves superior performance compared with the state-of-the-art methods.

II. Related Work

A. Visual Feature-Driven IVIF Methods

Visual feature-driven approaches constitute a core research paradigm in the IVIF domain. Their fundamental principle involves optimizing low-level visual representations via feature extraction and fusion strategies, thereby enhancing texture integrity, edge sharpness, and contrast balance of fused results [9], [20], [21], [22]. With the advent of deep learning, visual feature-driven methods enter an intelligent optimization era, where automated feature learning emerges as a central research focus.

Some works focus on network architecture optimization by enhancing the adaptation and extraction of bimodal features via a customized encoder–decoder architecture. The dual-branch feature extraction framework becomes a prevalent approach, as it accommodates the distinct characteristics

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of infrared (thermal radiation targets) and visible (textural details) modalities [10], [12], [13], [23], [24], [25], [26], [27]. Deng et al. [24] present MMDRFuse, adopting a lightweight distillation model combined with a dynamic refresh training strategy to achieve efficient cross-modal fusion using a minimal convolutional network. Zhao et al. [25] devise CDDFuse, which employs Restormer as the foundational feature extraction module, coupled with a dual-branch lite transformer INN for disentangling high-frequency and low-frequency features, and reinforces cross-modal alignment via correlation-driven losses. Zhang et al. [26] propose FSATFusion, which utilizes a dual-stream frequency–spatial attention transformer to capture multiscale modal features. Liu et al. [27] develop DCEvo, integrating a discriminator enhancer with a cross-dimensional embedding module and an evolutionary learning strategy to optimize discriminative feature extraction and enhance synergy with downstream tasks. Concurrently, techniques such as residual connections [28] and dense connections [29] are extensively embedded in architectural designs to mitigate feature degradation during propagation. However, these methods rely on predefined architectural priors and exhibit limited adaptability to the dynamic intermodal discrepancies between infrared and visible modalities.

Handcrafted loss function constraint methods aim to construct supervisory signals by manually defining mathematical constraints at the pixel, feature, and gradient levels, ensuring visual consistency between fused results and source images [14], [15], [30], [31], [32]. Hou et al. [30] employ a minimalist CNN framework, achieving noise suppression and detail preservation via a handcrafted mixed loss [M-structural similarity index measure (SSIM)+TV]. Tang et al. [31] adopt a standard U-Net architecture and adaptively balance thermal radiation from infrared images and texture details from visible images through a manually designed illumination-aware loss function. Bai et al. [32] propose a task-driven fusion method that designs a learnable fusion loss module within a metalearning framework, where the module optimizes loss parameters inversely based on downstream task performance to break through the fixed heuristic rules of traditional handcrafted losses. However, such manually designed loss functions still rely on the prior logic of pixel or structural similarity, exhibiting limited adaptability and constraint effectiveness in complex scenarios commonly encountered in IVIF, such as low-light conditions and high noise levels.

B. Downstream Task-Driven IVIF Methods

Downstream task-driven methods represent a pivotal paradigm in the evolution of IVIF from pure visual optimization to practical value orientation. Their core principle lies in transcending the traditional sequential framework of first fusion and then task execution, deeply coupling the fusion process with downstream tasks (e.g., target detection and semantic segmentation). Through task performance feedback, this approach enables reverse guidance for fusion feature learning, ensuring that fused results not only meet visual quality standards but also precisely align with task-specific input requirements [9], [33]. Cascaded optimization methods adopt a cascaded structure comprising a fusion network followed by a task network, incorporating downstream task losses into the fusion training process to achieve reverse optimization of fusion parameters based on task performance. Liu et al. [34] present an object-aware dual-adversarial learning framework, which leverages joint training of detection and fusion losses to enhance the preservation of object-critical features for detection tasks. Tang et al. [35] cascade a semantic segmentation network with a fusion network, utilizing segmentation loss to guide the fusion process in retaining semantically discriminative key features. While these methods are straightforward to implement and exhibit strong task specificity, they face challenges such as information loss during feature transmission and limited adaptability to multitask scenarios [9].

Feature interaction methods emphasize deep synergy between fusion and tasks at the feature level, thereby avoiding feature mismatch caused by simplistic cascading. Zhao et al. [36] introduce the MetaFusion framework, which incorporates a metalearning mechanism to simulate mutual enhancement between fusion and detection tasks via metafeature embedding. Liu et al. [37] design a multiinteractive feature learning network that adaptively adjusts feature interaction strength using dynamic weighting factors, thereby improving adaptability to multitask scenarios. However, the design of interaction mechanisms in such methods relies heavily on empirical heuristics, and the dynamic weight adjustment logic struggles to adapt to all complex scenarios, resulting in limitations in flexibility and generalization.

C. Text Semantic-Driven IVIF Methods

With the breakthrough advancements in VLMs [38], [39], the text-driven fusion paradigm emerges as a prominent research focus in recent years. Text-driven fusion methods leverage pretrained text language models to extract high-level semantic information, thereby enabling precise semantic control over the fusion process. Building upon pretrained text language models [38], [39], [40], researchers develope various text-guided fusion frameworks [16], [17], [41], [42], [43], [44]. Yi et al. [16] introduce the Text-IF framework, which achieves degradation-aware IVIF by leveraging semantic text guidance. It adapts to challenging scenarios such as low-light conditions and noise interference via text descriptions, thereby enabling customized preservation of complementary bimodal information. Zhu et al. [41] propose the TeSG framework that utilizes the pretrained BLIP VLM to extract word- and phrase-level semantic features. Through a dual-level (mask- and text-level) semantic optimization guidance mechanism, the framework achieves precise text-visual feature alignment and enhances cross-modal semantic anchoring. Zhang et al. [42] present the VGDCFusion framework. The framework employs VLMs as a degradation-aware bridge, generating modality-specific degradation prompts that are deeply integrated into the fusion pipeline, effectively adapting to semantic-guided fusion in diverse degradation scenarios. Wang et al. [43] propose the RIS-Fuse framework, which adopts a weakly supervised two-stage paradigm, leverages manifold similarity and spatial positional rules to generate instance-level pseudolabels, and achieves precise text-visual feature matching through its multimodal feature alignment module (MFA) for

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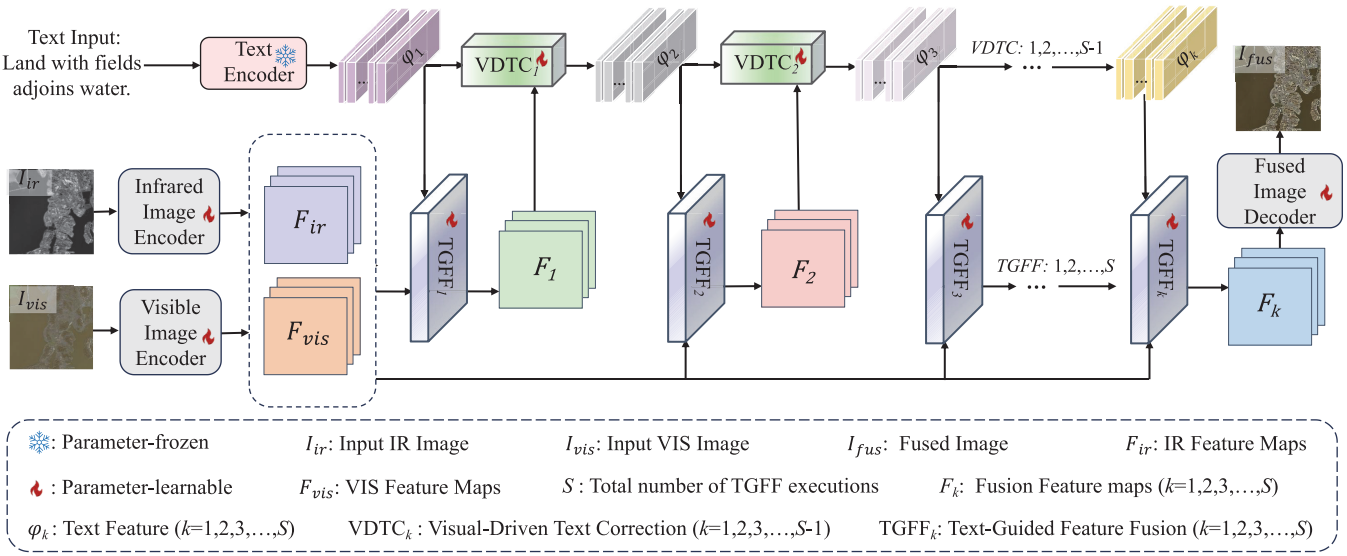


Fig. 3. Overall architecture of the end-to-end fusion framework VP-BIL. This framework regards infrared images, visible images, and remote sensing textual descriptions as input modalities. Features are sequentially extracted by a dual-branch U-Net and the CLIP encoder. Subsequently, multiple iterations between TGFF and VDTC achieve bidirectional alignment between textual and visual features, and the decoder eventually generates a high-quality fused image.

instance-level controllable semantic-guided fusion. Liu et al. [44] develop the PromptFusion method that utilizes text as the fusion constraint and employs both positive and negative prompts to construct contrastive learning, thereby significantly enhancing the feature representation of the fused images. Although these methods improve semantic controllability in fusion, they still suffer from dual limitations. First, insufficient semantic adaptation in complex scenarios leads to suboptimal fusion performance. In addition, the imbalance in the preservation ratio between visually guided semantically enhanced features and original visual features results in the loss of visual details [45].

In summary, current IVIF methods still exhibit multidimensional limitations: visual feature-driven approaches lack adaptive capability to dynamic modality discrepancies. Downstream task-driven methods demonstrate insufficient versatility for multitask adaptation. Text-guided fusion techniques require improvements in cross-modal semantic anchoring accuracy and scene generalization. To address these challenges, we propose a VP-BIL method. This approach employs a dual-U-Net architecture to separately extract infrared thermal radiation features and visible texture features. By integrating a cross-attention mechanism, it achieves precise text-visual semantic alignment. Furthermore, it introduces a residual fusion structure with learnable weights to balance original visual details and semantically enhanced features. Finally, it implements iterative bidirectional text-visual calibration to dynamically rectify textual description biases. This pathway ensures collaborative optimization of precise semantic alignment without deviation and complete preservation of original details in fusion results, effectively mitigating the triple limitations of current methods in semantic understanding, dynamic adaptation, and information retention.

III. Proposed Method

A. Overview

This article proposes a VP-BIL model, designed to address the unidirectional dependence bias and the semantic-visual information imbalance in existing text-guided image fusion methods. The model aims to meet the critical requirements of remote sensing applications for accurate preservation of thermal targets, complete retention of terrain textures, and unbiased semantic alignment [9]. VP-BIL adopts an end-to-end architecture, as shown in Fig. 3, with the overall workflow as follows. Two parameter-independent U-Net branches are employed to extract visual features, tailored to the characteristics of infrared and visible modalities, respectively. For text semantic encoding, the pretrained contrastive language-image pretraining (CLIP)-ViT-Base-Patch32 model [39] is adopted with its weights frozen to ensure stable cross-modal semantic extraction. Subsequently, a closed-loop optimization is achieved through the TGFF and VDTC, ultimately generating a high-quality fused image via a decoder. The overall mapping relationship is defined as

$$I_{\text{fus}} = \text{VP-BIL}\left(I_{\text{ir}}, I_{\text{vis}}, T; \boldsymbol{\theta}\right) \tag{1}$$

where $I_{\text{fus}} \in \mathbb{R}^{H\times W\times 1}$ denotes the single-channel grayscale fused image, $I_{\text{ir}} \in \mathbb{R}^{H\times W\times 1}$ and $I_{\text{vis}} \in \mathbb{R}^{H\times W\times 3}$ represent the single-channel infrared image and three-channel visible image, respectively, T denotes the text description of the remote sensing scene, $F(\cdot)$ signifies the overall mapping function of the proposed model, $\boldsymbol{\theta}$ represents the set of learnable parameters of the VP-BIL model.

B. Visual-Prompt Bidirectional Interactive Learning

VP-BIL is the key to performance enhancement in VP-BIL. Utilizing the deep features from the fifth layer of U-Net as the interactive medium, it adopts an empirically validated optimal configuration to construct a four-stage text-guided original

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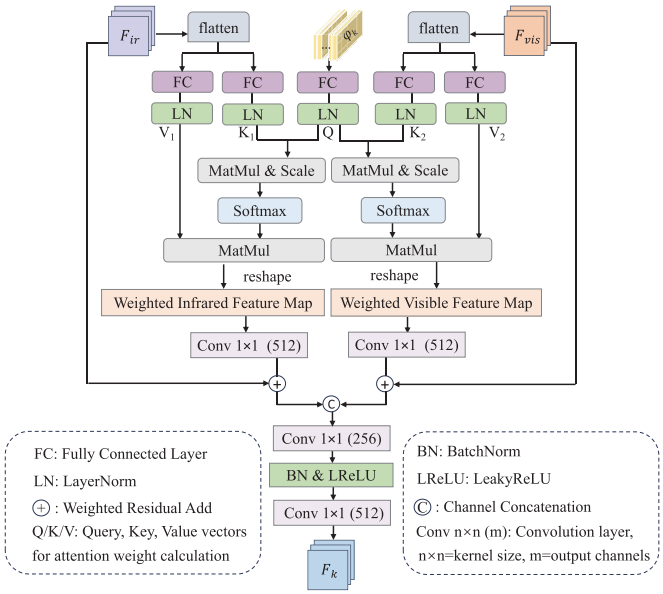


Fig. 4. Pipeline of TGFF. Infrared and visible features are mapped into a shared semantic space. Then cross-attention computation under the guidance of text features is conducted. Next, the original visual details with semantically enhanced features are balanced via adaptive residual fusion. Finally, TGFF outputs cross-modally aligned fusion feature maps.

visual feature fusion logic. In particular, after three rounds of TGFF-VDTC iterative interactive optimization, one final round of TGFF fusion generation is performed. This logic specifically addresses the critical issue of unidirectional textual dependence prevalent in conventional methods, effectively enhancing cross-modal semantic alignment and fine-grained visual details. Consequently, it successfully overcomes the core limitations inherent in existing text-guidance image fusion methods.

1) Text-Guided Feature Fusion: To mitigate potential noise interference arising from the large discrepancy between text and visual feature modalities, the TGFF ensures the rationality of text guidance through its core design: the module first projects text and bimodal visual features uniformly into a common semantic space to achieve alignment of dimensions and semantic distributions, and reduces noise induced by modality mismatch. Meanwhile, cross-attention only implements local guidance on regions corresponding to text semantics, avoiding the introduction of global noise.

Text-guided feature fusion (TGFF) serves as the core fusion module of the model, with its total number of invocations denoted as S (empirical studies have validated S = 4 as the optimal value). Here, k denotes the kth invocation of TGFF (k = 1, 2, 3, 4). Among these, k = 1, 2, and 3 correspond to the iterative optimization stages, while k = 4 signifies the final fused image generation stage. This module employs the textual features of the current iteration as semantic guidance to drive the fusion of the original visual features and incorporates a residual fusion structure with learnable weights. The design ensures precise alignment and fusion of infrared and visible features while preventing the loss of original visual information caused by semantic enhancement. The core process is illustrated in Fig. 4.

Following the flattening of infrared features F_{ir} and visible features F_{vis}, a fully connected layer is employed to map the features into a shared semantic space, thereby generating modality-specific key (K_1), value (V_1), and key (K_2), value (V_2) representations. This unifies feature dimensions and lays the foundation for effective cross-modal feature interaction. Concurrently, the text features φ_k ($k = 1, 2, 3, 4$) input at this stage are mapped into a query vector (Q) through a linear layer, ensuring strict feature dimensional consistency with that of K and V, thereby enabling the matrix multiplication operation in the subsequent cross-attention mechanism. Subsequently, layer normalization is applied individually to the generated K_1, V_1, K_2, V_2, and Q to enhance the stability of feature distributions and alleviate cross-modal misalignment caused by discrepancies in feature distributions across modalities.

On this basis, Q is taken as the sole semantic core to perform matrix multiplication with the keys corresponding to each modality, to calculate the cross-modal semantic similarity. The obtained similarity is simultaneously subjected to a scaling operation and temperature coefficient adjustment: the former stabilizes the numerical scale and prevents Softmax saturation, while the latter optimizes the attention weight distribution, mitigates semantic anchoring deviation, and further improves the accuracy of cross-modal semantic alignment [46]. Finally, Softmax normalization is conducted to generate attention weights highly consistent with textual semantics, thereby completing cross-modal semantic matching between text and vision, and laying a core computational foundation for the module to achieve fine-grained semantic alignment. The cross-attention calculation guided by cross-modal semantics is implemented independently for each modality, and the calculation of attention weights is given by the following equation:

$$\text{Attn}_1 = \text{SoftMax}\left(\frac{Q \cdot K_1^T}{\sqrt{d_k} \cdot \tau}\right) \tag{2}$$

$$\text{Attn}_2 = \text{SoftMax}\left(\frac{Q \cdot K_2^T}{\sqrt{d_k} \cdot \tau}\right) \tag{3}$$

where $1/((d_k)^{1/2})$ is the scaling factor (d_k denotes the dimension of the attention head), $\cdot$ represents matrix multiplication between the query and key vectors, τ is the temperature coefficient, and SoftMax normalizes the raw attention scores to the interval [0, 1], thereby enabling the probabilistic assignment of feature weights. $Attn_1$ and $Attn_2$ correspond to the cross-modal attention weights of the infrared modality and visible light modality, respectively. Based on the aforementioned attention weights, weighted aggregation is separately implemented on the values of the corresponding modalities, which realizes adaptive screening and enhancement of visual features in accordance with textual semantic relevance, whereby valid information with high semantic correlation with texts is preserved, and irrelevant interference is suppressed to output the semantically enhanced feature sequences of infrared and visible modalities. This process accurately embeds textual semantics into visual features, establishes stable semantic correlations between the two modalities, and achieves cross-modal fine-grained semantic alignment. Nevertheless, the above alignment mechanism overemphasizes semantic

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correlation characteristics while weakening original inherent visual details. To address this issue, the enhanced feature sequences are then back-projected via a 1×1 convolution to the original visual feature dimensions and adaptively fused with F_{ir} and F_{vis} via a residual fusion structure, which dynamically balances the strength of semantic guidance and the preservation of original visual details to prevent the loss of original visual details during the semantic enhancement process, as shown in the following equation:

$$F_k^{\mathrm{IR}} = \alpha_1 \cdot F_{\mathrm{IR}} + \alpha_2 \cdot \mathrm{ProjBack}\left(\mathrm{Attn}_1 \odot \mathrm{V}_1\right) \tag{4}$$

$$F_k^{\mathrm{VIS}} = \alpha_1 \cdot F_{\mathrm{VIS}} + \alpha_2 \cdot \mathrm{ProjBack}\left(\mathrm{Attn}_2 \odot \mathrm{V}_2\right) \tag{5}$$

where α_1 and α_2 denote learnable residual fusion weights, ProjBack represents the feature back-projection operation. $\odot$ signifies elementwise multiplication. F_k^{IR} and F_k^{VIS} denote the infrared and visible cross-attention features derived from the kth invocation of TGFF.

Concatenation of F_k^{IR} and F_k^{VIS} along the channel dimension yields a high-dimensional feature space with intrinsic redundancy. To address this issue, we employ a dual 1×1 convolutional layer structure that adopts a progressive workflow consisting of dimension reduction, nonlinear enhancement, and dimension restoration, thereby avoiding information loss induced by single-step dimensionality reduction. In particular, the first 1×1 convolutional layer compresses the channel dimension to focus on core features, followed by BatchNorm to stabilize the feature distribution and mitigate internal covariate shift. A LeakyReLU activation function [47] is then incorporated to introduce nonlinearity, thereby enhancing feature representation capability. Subsequently, the second 1×1 convolutional layer conducts deep feature fusion and restores the channel dimension to match the required dimension of subsequent modules, ultimately outputting the cross-modal fusion feature F_k.

2) Visual-Driven Text Correction: Most existing multimodal alignment strategies correct text via approaches such as cross-attention hard alignment and direct feature replacement under visual guidance, which tend to destroy the structural integrity of text features, cause semantic distortion or excessive correction, and can hardly meet alignment requirements in complex scenarios. On the contrary, this article proposes a targeted lightweight correction mechanism: a scaling-shift framework is adopted to conduct fine-grained lightweight adjustment on the distribution of text semantics, which realizes the alignment between text features and visual semantics in complex scenarios without altering the inherent structure of text features, and fundamentally avoids feature destruction and semantic distortion caused by hard correction.

The visual-driven text correction (VDTC) mechanism, based on the fused visual feature F_k output by TGFF, is activated exclusively during the iterative interaction phase ($k = 1, 2, 3$) to rectify semantic deviations in the textual features, thereby compensating for the lack of fine-grained terrain details in remote sensing text descriptions. This module is omitted during the final generation stage ($k = 4$). The core workflow is illustrated in Fig. 5. To extract high-level semantic features that encompass both the global structural information and key target representations within remote sensing scenes, the global

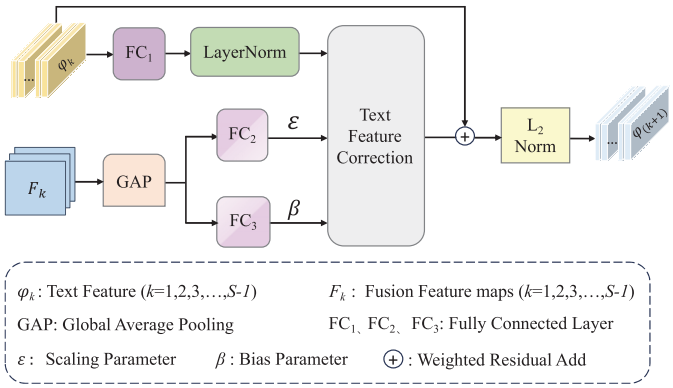


Fig. 5. Pipeline of VDTC. It extracts global visual semantics from the fused features, performs text feature correction (scale-offset transformations) and residual fusion to dynamically adjust the semantic deviation of the current text features, and finally outputs the optimized text features for subsequent interactive iterations.

average pooling (GAP) operation [48] is applied to the feature maps F_k ($k = 1, 2, 3$). This operation computes the spatialwise mean value of all elements along the spatial dimensions for each channel of F_k independently, thereby integrating global semantic information while reducing feature dimensionality, ultimately yielding the global visual semantic vector V_{global}, where the output spatial dimension of each channel of this vector is 1×1, corresponding to a single scalar value. The mathematical formulation of this operation is given by the following equation:

$$V_{\mathrm{global},c} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} F_k(c, i, j) \tag{6}$$

where $V_{\mathrm{global},c}$ denotes the scalar element corresponding to the cth channel of the global visual semantic vector V_{global}. The variable c is the channel index of the input feature map F_k with $c \in [1, C]$, where C denotes the total number of channels of F_k. The indices i and j denote the spatial height and width indices of F_k with $i \in [1, H]$ and $j \in [1, W]$, with H and W representing the spatial height and width dimensions of F_k. In addition, $F_k(c, i, j)$ denotes the element value at the cth channel and spatial position (i, j) in F_k, $\sum_{i=1}^{H} \sum_{j=1}^{W}$ denotes the summation operator, and $1/(H \times W)$ is the averaging coefficient.

We design two fully independent fully connected layers (both without bias terms) to convert the global visual semantics into modulation parameters for textual feature calibration, thereby mitigating textual semantic bias to align textual features more closely with the visual semantic information of the original images. These layers project V_{global} into the scaling vector ε and offset vector β required for textual correction [48], respectively. The advantage of this design lies in preserving the visual semantic purity of the modulation parameters. The specific formulations are given in the equations

$$\varepsilon = \mathbf{W}_{\varepsilon} \cdot V_{\mathrm{global}} \tag{7}$$

$$\beta = \mathbf{W}_{\beta} \cdot V_{\mathrm{global}} \tag{8}$$

where $\mathbf{W}_{\varepsilon}, \mathbf{W}_{\beta} \in \mathbb{R}^{512 \times 512}$ denote the learnable projection matrices.

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Meanwhile, to achieve dimensional consistency between the textual features and ε, β while enhancing the stability of the textual features' semantic representations, we apply a linear projection to the textual features φ_k of the current iteration and normalize them via layer normalization (LayerNorm) to regularize the feature distribution. We then apply the previously introduced ε and β to the normalized textual features. ε performs channelwise amplification or attenuation on them, whereas β compensates for the global shift in semantic distribution. Through this scale-shift mechanism, we dynamically adjust the textual semantic distribution to align with the visual semantics of the remote sensing scene, ultimately yielding the preliminarily adjusted features $\varphi_{\text{adjusted}}$. The specific computation is given by the following equation:

$$\varphi_{\text{adjusted}} = \text{LayerNorm}\left(\mathbf{W}_t \cdot \varphi_k\right) \odot \varepsilon + \beta \tag{9}$$

where $\mathbf{W}_t \in \mathbb{R}^{512\times512}$ denotes the learnable linear projection matrix, and LayerNorm refers to the layer normalization operation. To retain the core textual semantics and enable iterative optimization during the correction process, we adopt a residual fusion strategy to fuse the preliminarily adjusted features $\varphi_{\text{adjusted}}$ with the input textual features φ_k of the current iteration. This strategy retains the original textual information while mitigating semantic bias. Subsequently, L2 normalization is applied to the fused features to further regularize the feature distribution, resulting in the textual features $\varphi^{(k+1)}$ for TGFF input in the next iteration, as formulated in the following equation:

$$\varphi_{(k+1)} = \text{L2Norm}\left(\varphi_k + \gamma \cdot \varphi_{\text{adjusted}}\right) \tag{10}$$

where γ denotes the learnable initial correction weight that ensures the correction intensity is appropriately calibrated to the requirements of remote sensing scenarios, and L2Norm refers to the L2 normalization operation.

3) Iterative Interaction Mechanism: The three-round iterative interaction serves as the core carrier of the bidirectional interaction mechanism, in which textual features are continuously updated throughout the iterations while consistently guiding the fusion of original visual features. Its detailed workflow is described as follows: in the first round, the initial textual feature φ_1 output from the CLIP encoder drives the TGFF to generate the fused feature F_1. F_1 is then fed into the VDTC, generating the updated textual feature φ_2. With φ_2 as the updated semantic guidance, the TGFF generates a higher precision fused feature F_2 in the second round, which is subsequently fed into the VDTC to produce φ_3. In the third round, φ_3 drives the TGFF to produce F_3, which is then fed into the VDTC to output the optimized textual feature φ_4. Finally, with φ_4 as the semantic guidance, the TGFF generates the final fused feature F_4, which is fed into the decoder to reconstruct a high-quality fused image.

C. Optimization

To ensure consistency in luminance, contrast, and structural characteristics between the fused image and the source images, and to meet the requirements of preserving ground object structures in remote sensing scenarios, we adopt the SSIM [11] loss to construct the training constraint. With the fused image I_{fus}, the infrared image I_{ir}, and the luminance channel of the visible image $I_{\text{vis}}^{\text{lum}}$ as References, the total loss function is defined as shown in the following equation:

$$L_{\text{total}} = \left(1 - \text{SSIM}(I_{\text{fus}}, I_{\text{ir}})\right) + \left(1 - \text{SSIM}(I_{\text{fus}}, I_{\text{vis}}^{\text{lum}})\right). \tag{11}$$

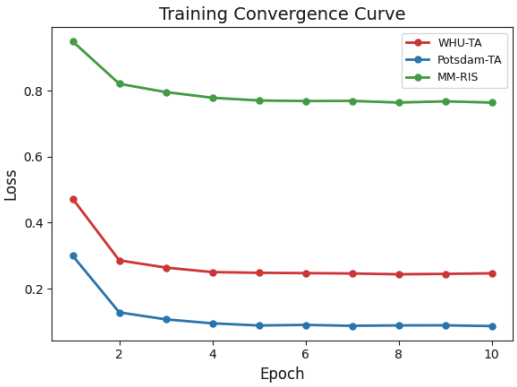


Fig. 6. Training loss convergence curves of the model on WHU-TA, Potsdam-TA, and MM-RIS datasets.

IV. Experiments

A. Experimental Setting

1) Implementation Details: Our method is implemented in PyTorch and trained on an NVIDIA GeForce RTX 4090 GPU with 24 GB of VRAM. For training, we utilize the Adam optimizer with an initial learning rate of 10^{-5} and a weight decay of 10^{-5}, conducting 5 training epochs with a batch size of 2. Beyond this epoch, the model fusion metric declines due to overfitting, indicating that 5 epochs of training achieve the optimal performance of the model. To further quantitatively verify the convergence and stability of the training process, Fig. 6 presents the training loss convergence curves of the model on the custom datasets WHU-TA, Potsdam-TA, and the general dataset MM-RIS [49]. It can be observed from the curves that the loss of the model on all datasets converges within five epochs and enters a stable plateau without obvious oscillation, indicating that five training epochs are sufficient for the model to reach the optimal convergence state, and the training process is stable and reliable. Input images undergo standardized preprocessing. Visible RGB images are converted to the YCbCr color space, and only the Y channel is retained, while single-channel infrared images are used directly without additional preprocessing. Both types of images are directly used with their native resolution of 512×512 pixels to match the model input dimensions, and normalized to the range [0, 1] via min–max normalization. The model employs a U-Net backbone with the number of base channels configured to 64. The number of interactive iteration rounds between the TGFF and VDTC is set to a fixed value of 3. The core hyperparameters of the model are summarized as follows: $\tau = 0.6$. $d_k = 240$. $\alpha_1 = 0.1$. $\alpha_2 = 0.9$. $\gamma = 2.0$. The initial values of the learnable weighting coefficients for shallow-to-middle layer infrared features are all initialized to 0.8, and no additional weighting is imposed on visible features of the

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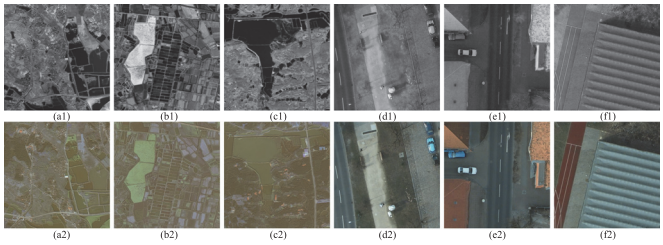


Fig. 7. Example pairs of infrared and visible images with corresponding textual descriptions from WHU-TA and Potsdam-TA. The upper row (a1)–(f1) shows the infrared images, and the lower row (a2)–(f2) shows the corresponding visible images. The semantic text descriptions for each image pair are listed as follows. (a) Fields on the land have roads and buildings. (b) There are roads passing through fields. (c) Roads cross land with fields adjacent to water. (d) People and vehicles are by trees. (e) Cars are parked by buildings. (f) Trees are beside the building and the playground.

corresponding layers. The computational complexity of our model is 108.54 GFLOPs, and the average inference time for a single image is 10.98 ms.

2) Datasets: We annotate two widely used remote sensing datasets, namely WHU-TA and Potsdam-TA, which are designed for the text-guided IVIF task. The datasets contain a total of 1600 image pairs, covering typical remote sensing scenes including urban landscapes, terrain, and vegetation. In particular, WHU-TA is derived from the publicly available WHU dataset [50], consisting of 800 representative image pairs selected from the original dataset (500 for training and 300 for testing). Potsdam-TA is built upon the publicly available Potsdam dataset [51], consisting of 800 image pairs (500 for training, 300 for testing).

As illustrated in Fig. 7, several typical image–text paired examples are provided to demonstrate the overall construction of the dataset. To ensure the standardization, consistency, and reliability of annotations, eight professional annotators from our research group jointly completed all sample annotation work in this study. Prior to annotation, the team jointly formulated unified annotation guidelines: all annotations must contain the core information of target categories and spatial positional relationships, adopt objective declarative sentences, and avoid subjective evaluation and redundant modifiers. For spatial description, precise expressions suited for remote sensing scenarios such as adjacent, surrounding, penetrating, and lateral are adopted to clearly define the relative positions between ground objects. In view of the scene characteristics of the two datasets, the annotation content was designed with different emphases. WHU-TA focuses on rural natural and farmland landscapes, with key annotations on rural terrain-related targets such as landforms, vegetation, farmlands, water bodies, and roads. On the contrary, Potsdam-TA focuses on urban building areas, and its annotation centers on typical urban scene targets such as architectural components, urban roads, urban vegetation, vehicles, and pedestrians. During the annotation process, eight annotators simultaneously observed and identified salient objects for each image pair. The annotated content was incorporated into the final version only when all members reached a complete consensus on the description of target categories and positions. If any divergence existed, the final result was confirmed after collective review and calibration by all members to guarantee consistent annotation results.

The final annotated texts are characterized by conciseness and uniformity. All descriptions adopt a single-sentence structure, with the number of words ranging from three to ten and an average of 6.2 words. Such texts can not only fully convey core semantics but also satisfy the input requirements of text encoders. In terms of vocabulary, the texts cover eleven categories of core remote sensing ground objects and five categories of commonly used orientation expressions, with 426 and 19 unique entries after deduplication, respectively. Free from redundant repetition, these vocabularies can fully support the model in learning diverse text-visual semantic correspondences. This full-consensus-based annotation mode inherently guarantees data reliability and consistency, and provides high-quality data support for subsequent model training. The WHU-TA and Potsdam-TA datasets used in this work are publicly available at our GitHub repository: https://github.com/didi915/VP-BIL-for-Remote-Sensing-Infrared-and-Visible-Image-Fusion

To further objectively verify the generalization capability of the proposed model, generalization tests were carried out on the public universal text–image paired dataset MM-RIS. As an independent public standard dataset, MM-RIS is free from any customized textual annotation in this study, which can rigorously verify the generalization performance of the model in nondedicated text scenarios.

3) Evaluation Metrics: This study employs six evaluation metrics to assess the information richness and visual quality of fused images, where higher values indicate superior fusion performance. In particular, information entropy (EN) [1] quantifies the information-carrying capacity of fused images, standard deviation (SD) [52] reflects the dispersion degree of the grayscale distribution in fused images (corresponding to contrast), spatial frequency (SF) [53] measures the richness of fine-grained structural details in fused images, average gradient (AG) [54] evaluates the overall clarity and edge sharpness of fused images, structural similarity-based correlation coefficient (SCD) [55] characterizes the structural correlation fidelity between fused images and source images, and visual information fidelity for fusion (VIF) [52] assesses the visual restoration effect of fusion results from the perspective of human visual perception.

B. Comparison With SOTA Methods

We compare our method with eight state-of-the-art (SOTA) fusion approaches on the WHU-TA, Potsdam-TA, and MM-RIS datasets. The compared approaches include: GFF [56], U2Fusion [10], LRRNet [57], MMDRFuse [24], SAGE [58], TextFusion [18], Text-IF [16], and FILM [17]. Among them, GFF is a conventional image fusion method based on guided filtering. It splits images into base layers and detail layers and completes fusion through weighted combination.

1) Qualitative Comparison: We conduct a qualitative comparative analysis of the proposed VP-BIL method against eight SOTA fusion approaches on the WHU-TA and Potsdam-TA

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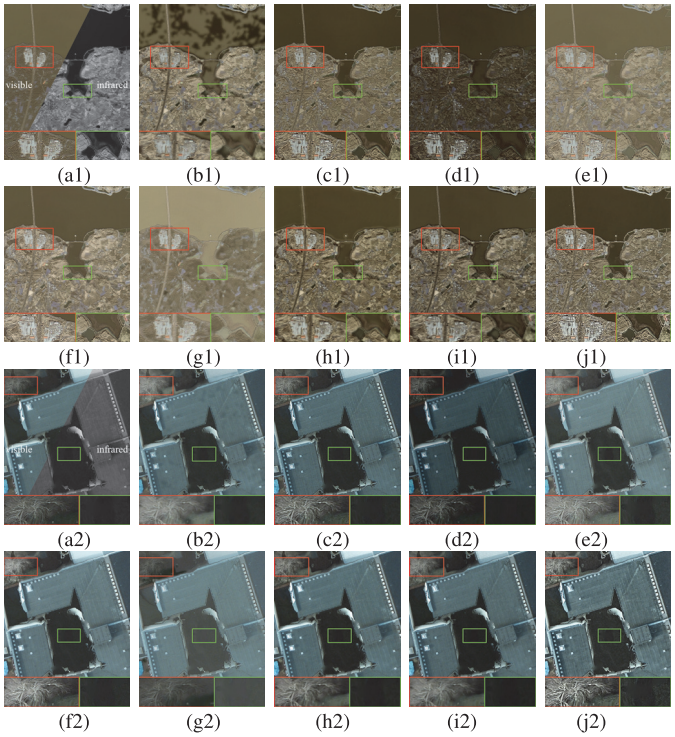


Fig. 8. Visual comparison of different fusion methods on WHU-TA (first group) and Potsdam-TA (second group), respectively. (a1) and (a2) Source image. (b1) and (b2) GFF. (c1) and (c2) U2Fusion. (d1) and (d2) LRRNet. (e1) and (e2) MMDRFuse. (f1) and (f2) SAGE. (g1) and (g2) TextFusion. (h1) and (h2) Text-IF. (i1) and (i2) FILM. (j1) and (j2) Ours.

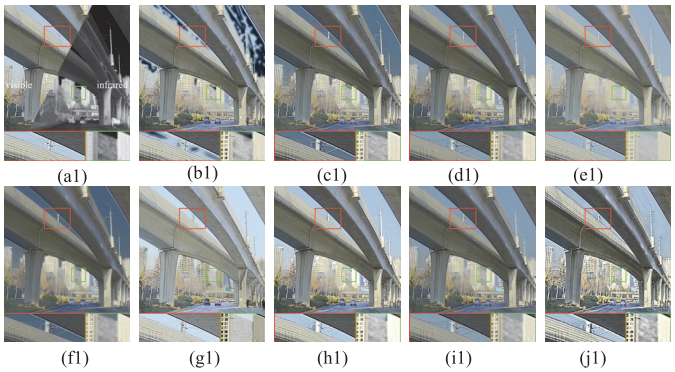


Fig. 9. Visual comparison of different fusion methods on the MM-RIS dataset. (a1) Source image. (b1) GFF. (c1) U2Fusion. (d1) LRRNet. (e1) MMDRFuse. (f1) SAGE. (g1) TextFusion. (h1) Text-IF. (i1) FILM. (j1) Ours.

datasets. The comparative results are illustrated in Fig. 8. Generalization evaluation of the proposed model is carried out on the MM-RIS dataset, and the relevant experimental results are presented in Fig. 9. Experimental results demonstrate that all existing methods fail to adequately achieve three objectives simultaneously: highlighting the saliency of infrared thermal targets, preserving visible texture details, and recovering semantic features, with remarkable deficiencies observed in the key annotated regions. In particular, GFF and LRRNet both suffer from the problem of an unbalanced infrared-visible feature fusion ratio. GFF exhibits distinct black artifacts distributed throughout the entire image, where the textures of contiguous building clusters are blurred and faded, and the details of tree crowns adjacent to building roofs are lost, indicating its poor adaptability in the allocation of fusion weights between thermal targets and visible textures. Although LRRNet can preserve the overall contours of ground objects, the saliency of thermal targets in shadowed depressions and dark grassland shadow areas is insufficient, which easily leads to feature aliasing with the background. U2Fusion, Text-IF, TextFusion, and FILM all fail to adapt to the multiscale characteristics of remote sensing scenarios, with limited capability in capturing fine-grained details. U2Fusion has weak clarity at the light–dark boundary of shadowed depressions, where the boundaries between dark grassland shadows and surrounding ground objects are blurred, along with limited detail restoration. TextFusion demonstrates a prominent global oversmoothing effect, which suppresses the overall image detail information: the texture details of contiguous building clusters degenerate, the texture features of vegetation are substantially attenuated and lost, and the boundaries of dark grassland shadow areas are diffused with insufficient edge sharpness; it also shows a prominent problem of scene detail loss. Text-IF exhibits weakened texture layering in contiguous building clusters, insufficient discriminability of local tree details, ambiguous demarcation in dark grassland shadow areas, and poor retention of refined structural features. FILM has weak texture contrast in contiguous building clusters with indistinct detail levels, and its depiction of light–dark levels in dark grassland shadows is inadequate; it has limited fine representation capability for multiple scenarios. MMDRFuse and SAGE have obvious defects in contrast balance and edge transition. The overall fused image of MMDRFuse has low contrast, poor feature discriminability for contiguous building clusters, shadowed depressions, and vegetation clusters, excessively soft edge transitions, and insufficient dual-modal information fusion. SAGE exhibits unnatural edge transitions with indistinct boundaries between the core region and surrounding ground objects, and partial scenes suffer from color deviation and a dark, dull visual tone.

For scenarios in the MM-RIS dataset, existing comparative fusion methods are prone to problems such as cable textures being covered by black spot noise, decreased edge discriminability, insufficient contour regularity, and insufficient edge sharpness when processing overhead cable targets. Meanwhile, all methods achieve poor performance in representing building gap shadow areas, with undesirable manifestations including dull and monotonous overall tone, inadequate depiction of shadow layering, and insufficient presentation of internal details.

By comparison, the proposed VP-BIL approach mitigates the aforementioned deficiencies through a bidirectional interaction mechanism of text-guided feature fusion and vision-corrected textual bias. It exhibits superior fusion performance characterized by rich and clearly distinguishable textures of continuous building clusters, well-defined boundaries of depression shadow regions, sharp fine-grained contours of vegetation canopies, intact morphological contours of dark grassland shadow regions, continuous and regular overhead cable lines, well-layered shadows in building gaps, and prominently highlighted infrared thermal targets. This

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TABLE I

QUANTITATIVE COMPARISON OF IMAGE FUSION WITH SOTA METHODS ON WHU-TA AND POTSDAM-TA. THE BEST RESULTS ARE HIGHLIGHTED IN RED AND THE SECOND RESULTS ARE HIGHLIGHTED IN BLUE

Method	WHU-TA						Potsdam-TA					
	EN	SD	SF	AG	SCD	VIF	EN	SD	SF	AG	SCD	VIF
GFF [56]	6.3782	25.6077	18.0746	6.6227	0.8521	1.1034	6.6702	30.9013	11.5608	4.0909	0.7086	1.4582
U2Fusion [10]	6.3881	29.9128	24.0688	9.5820	1.1260	0.9857	6.7505	34.0570	16.0844	6.0845	0.8190	1.4587
LRRNet [57]	6.2728	22.8188	18.9064	6.8130	0.4342	0.7263	6.7450	33.3752	11.8059	4.2118	0.6675	1.1475
MMDRFuse [24]	6.1191	30.6080	24.7548	9.0988	1.1889	1.0020	6.6955	34.3675	14.0380	5.0264	0.7993	1.4676
SAGE [58]	6.4409	33.9694	23.0393	8.8178	1.2891	0.9868	6.9683	41.6578	14.0803	5.1487	1.4884	1.5111
TextFusion [18]	5.4379	30.0474	20.8688	8.4721	0.6969	0.3564	6.2275	38.6117	17.6917	6.2589	0.7358	0.7602
Text-IF [16]	6.5763	26.9441	23.2151	8.4119	1.4256	1.1617	6.8897	35.6755	14.4531	5.1415	1.5340	1.5812
FILM [17]	6.6912	28.9208	21.8454	8.1216	1.4938	1.0983	6.8621	34.3163	12.7666	4.5669	1.3593	1.4722
Ours	6.9613	35.6344	39.2144	13.7745	1.6242	1.1057	7.2130	41.9815	25.4023	9.4187	1.3890	1.5875

TABLE II

QUANTITATIVE COMPARISON OF IMAGE FUSION WITH SOTA METHODS ON MM-RIS. THE BEST RESULTS ARE HIGHLIGHTED IN RED AND THE SECOND RESULTS ARE HIGHLIGHTED IN BLUE

Method	MM-RIS					
	EN	SD	SF	AG	SCD	VIF
GFF [56]	6.7348	38.2183	13.1675	4.3645	0.9874	0.6774
U2Fusion [10]	6.2149	31.8429	11.1018	3.6075	1.4188	0.6288
LRRNet [57]	6.3283	29.7567	9.1726	2.8073	0.9718	0.5706
MMDRFuse [24]	6.8159	39.2208	13.3449	4.0211	1.5772	0.7023
SAGE [58]	6.2903	26.6765	6.7318	2.1922	1.2780	0.6131
TextFusion [18]	6.7704	39.5460	14.8984	4.4160	0.8437	0.6707
Text-IF [16]	6.6460	32.4798	12.6139	3.7688	1.1697	0.7025
FILM [17]	6.8511	41.1889	13.3814	4.1111	1.4802	0.9524
Ours	6.9621	48.7283	22.2368	6.7850	1.5293	0.7099

method achieves a balanced fusion of infrared and visible dual-modal information, with significantly improved spatial–visual coherence and semantic consistency that outperform all competing methods. Consequently, it is well-aligned with the fusion requirements of complex scenarios.

2) Quantitative Comparison: Table I presents the quantitative evaluation results across WHU-TA and Potsdam-TA datasets using six metrics. Table II presents the quantitative evaluation results of model generalization capability tests carried out on the MM-RIS dataset. The proposed VP-BIL model achieves exceptional performance, outperforming all competing SOTA methods on WHU-TA, Potsdam-TA, and MM-RIS datasets. In particular, four evaluation metrics attain optimal values in various experimental scenarios, and the remaining two metrics maintain satisfactory comprehensive performance. The superior performance of the EN and SD metrics indicates that the generated fused images retain abundant global information, possess preferable discrete grayscale distribution characteristics, accurately integrate complementary information from multimodal remote sensing, and are free of feature degradation or redundant information accumulation, thus achieving comprehensive and compact fusion of multi-source information. The leading performance of the SF and AG metrics reflects that the fused images have more complete fine-grained details, sharper edge contours of ground objects, more distinct hierarchical features of topographic textures, and clearer boundary definition of target edges. The visual characteristics of full texture details and sharp edge contours presented in the fusion results are mutually corroborated with the excellent numerical performance of the SF and AG metrics. This outstanding performance originates from the core constraint of SSIM loss on structure, texture, and edge fidelity, which drives the model to naturally restore authentic fine-grained visual features of the source images. The SCD and VIF metrics attain a moderately high overall performance level, revealing satisfactory structural correlation and consistency of fused images with source images, acceptable visual information fidelity, and favorable structural rationality as well as visual naturalness of fusion results.

Apart from various deep learning-based approaches, a representative conventional nondeep learning fusion algorithm, GFF, is additionally adopted for comparative analysis. The proposed method achieves superior overall performance against GFF. Taking the Potsdam-TA dataset as an instance, the performance gains are 0.5428 for EN, 11.0802 for SD, 13.8415 for SF, 5.3278 for AG, 0.6804 for SCD, and 0.1293 for VIF. The experimental results demonstrate that the presented algorithm possesses distinct superiority over both deep learning and conventional nondeep learning counterparts.

Under the test scenarios of the MM-RIS dataset, the proposed method VP-BIL achieves the optimal results on all four metrics of EN, SD, SF, and AG, and obtains suboptimal performance on the two metrics of SCD and VIF.

In summary, the proposed VP-BIL model demonstrates significant metric advantages across three typical complex scenarios (i.e., terrain hydrography, building-road networks, and general real scenes), exhibiting its outstanding fusion performance and robust generalization capability and thus producing higher quality fused results.

C. Segmentation Results

To validate the effectiveness of our fusion results in downstream tasks, we conduct a comprehensive analysis on object segmentation using mainstream semantic segmentation networks. In particular, we train the segmentation network on the fusion images from the training set and evaluate its segmentation performance on the fusion images from the testing set. All semantic segmentation models are retrained under unified

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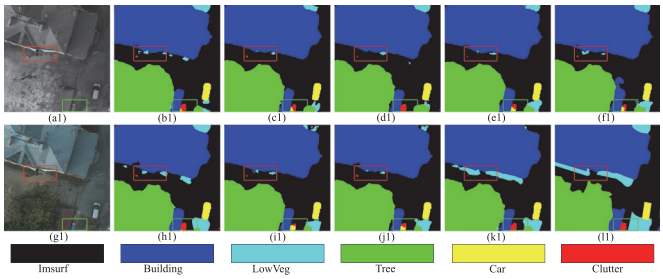


Fig. 10. (a1) Infrared image. (b1) GFF. (c1) U2Fusion. (d1) LRRNet. (e1) MMDRFuse. (f1) SAGE. (g1) Visible image. (h1) TextFusion. (i1) Text-IF. (j1) FILM. (k1) Ours. (l1) Ground Truth. Qualitative comparison of different fusion methods for the segmentation task on the Potsdam-TA dataset.

TABLE III

QUANTITATIVE SEMANTIC SEGMENTATION RESULTS OF DIFFERENT METHODS ON POTSDAM-TA DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN RED AND THE SECOND RESULTS ARE HIGHLIGHTED IN BLUE

Method	Potsdam-TA		
	mAcc	mIoU	mPA
GFF [56]	88.91	73.91	85.24
U2Fusion [10]	89.83	75.79	86.65
LRRNet [57]	89.60	75.22	86.53
MMDRFuse [24]	89.36	74.98	86.50
SAGE [58]	89.40	74.60	85.22
TextFusion [18]	88.64	73.64	84.98
Text-IF [16]	89.29	74.61	86.02
FILM [17]	88.90	74.27	85.76
Ours	90.27	77.13	86.73

experimental environments to maintain consistent hyperparameter configurations, so as to ensure fair comparison.

1) Qualitative Comparison: Visualized segmentation results on the Potsdam-TA dataset are depicted in Fig. 10, where we can clearly observe two remarkable advantages of our method. On the one hand, thanks to the sharper edge details in fusion results, our method can acquire more distinct segmentation edges, such as contour boundaries of low-vegetation regions and edge outlines of vehicle objects. On the other hand, due to the fact that fusion results carry richer object semantic information, our method is capable of distinguishing confused objects effectively, including junction areas between low vegetation and background, as well as ground objects with similar features such as vehicles and interference features. Our method utilizes segmentation-related constraints to obtain features with complete information representations, thus enabling fusion results to achieve satisfactory performance in segmentation tasks.

2) Quantitative Comparison: Three metrics are utilized to quantitatively evaluate the performance in downstream segmentation tasks, including mean accuracy (mAcc), mean intersection over union (mIoU), and mean pixel accuracy (mPA). Higher metric values represent better segmentation performance achieved by fused images. Table III reports the quantitative segmentation results of different comparison methods on the Potsdam-TA dataset. Our method achieves

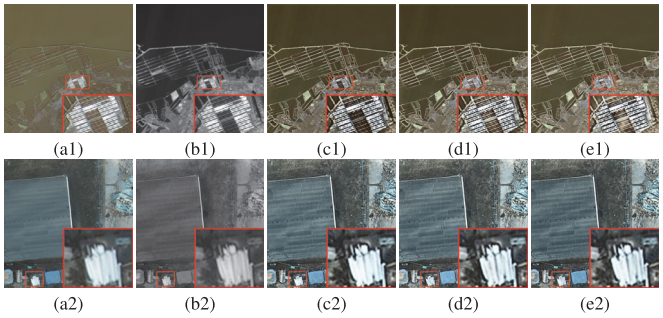


Fig. 11. (a1) and (a2) Visible image. (b1) and (b2) Infrared image. (c1) and (c2) VP-BIL-2iter. (d1) and (d2) Ours:VP-BIL-3iter. (e1) and (e2) VP-BIL-4iter. Visual comparison of different interaction iteration times for the VP-BIL model on WHU-TA (first group) and Potsdam-TA (second group). The VP-BIL-3iter precisely balances thermal target saliency and terrain detail integrity without visual distortions such as darkening and overexposure while ensuring superior fusion quality and high computational efficiency, serving as the optimal configuration for bidirectional interaction.

optimal values on three metrics, including mean accuracy (mAcc), mIoU, and mPA, and it is obviously superior to all other competitors. In particular, compared with the second-best method U2Fusion, our method outperforms it with mAcc increased by 0.44, mIoU increased by 1.34, and mPA increased by 0.08. Since the fused images generated by our method can contain more salient texture and edge information, the adequate information representation further enhances their performance in downstream segmentation tasks.

D. Ablation Study

To assess the contribution of each core module to the VP-BIL model's overall performance, we conduct ablation experiments on the WHU-TA and Potsdam-TA datasets. All variants strictly adhere to the single-variable control principle, maintaining identical training hyperparameters, feature extraction architecture, text encoder, and loss function as the original model to ensure comparative fairness. In the quantitative results, all metrics are evaluated according to the criterion that higher values indicate better performance.

1) Impact of the Interactive Iteration Times: To validate the optimal configuration of the bidirectional interactive iteration count for TGFF-VDTC, we design two variants with two (VP-BIL-2iter) and four (VP-BIL-4iter) interactive iterations, and compare them with the baseline VP-BIL with three iterations (VP-BIL-3iter). Quantitative results are tabulated in Table IV, whereas qualitative comparisons are visualized in Fig. 11. The results indicate that VP-BIL-2iter exhibits a significant degradation in key metrics: the SD, SF, and SCD decrease by 5.08%, 4.19%, and 3.52%, respectively. Its fused images exhibit severe quality degradation, including localized darkening and black artifacts in the grid-patterned parcel region where textural details are completely obscured by excessive darkening, as well as blurred contour boundaries with insufficient fine-grained details in the dark shadow regions of the desktop. These observations suggest that an inadequate iteration count hinders the complete convergence of cross-modal semantic alignment, thus impeding the effective operation of the bidirectional optimization mech-

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TABLE IV

QUANTITATIVE COMPARISON OF DIFFERENT INTERACTION ITERATION TIMES FOR THE VP-BIL MODEL ON WHU-TA AND POTSDAM-TA. THE BEST RESULTS ARE HIGHLIGHTED IN RED AND THE SECOND RESULTS ARE HIGHLIGHTED IN BLUE

Method	WHU-TA						Potsdam-TA					
	EN	SD	SF	AG	SCD	VIF	EN	SD	SF	AG	SCD	VIF
VP-BIL-2iter	6.8875	33.5380	37.3709	13.1637	1.5287	1.0925	7.1547	40.1828	24.4677	9.0428	1.3319	1.5887
VP-BIL-3iter	6.9613	35.6344	39.2144	13.7745	1.6242	1.1057	7.2130	41.9815	25.4023	9.4187	1.3890	1.5875
VP-BIL-4iter	6.9439	35.1920	37.8847	13.3018	1.6329	1.1008	7.1469	40.4276	25.1774	9.2566	1.3101	1.5712

TABLE V

QUANTITATIVE COMPARISON OF TGFF VARIANTS ON WHU-TA AND POTSDAM-TA. THE BEST RESULTS ARE HIGHLIGHTED IN RED AND THE SECOND RESULTS ARE HIGHLIGHTED IN BLUE

Method	WHU-TA						Potsdam-TA					
	EN	SD	SF	AG	SCD	VIF	EN	SD	SF	AG	SCD	VIF
TGFF-w/o-FusionRes	6.9364	34.7497	37.9647	13.3375	1.5949	1.0994	7.1118	39.7530	24.2208	8.8889	1.2892	1.5695
TGFF-SA-TCA	6.8992	34.1088	37.2467	12.9941	1.5748	1.0802	7.1084	39.4084	23.8628	8.7576	1.3211	1.5703
VP-BIL	6.9613	35.6344	39.2144	13.7745	1.6242	1.1057	7.2130	41.9815	25.4023	9.4187	1.3890	1.5875

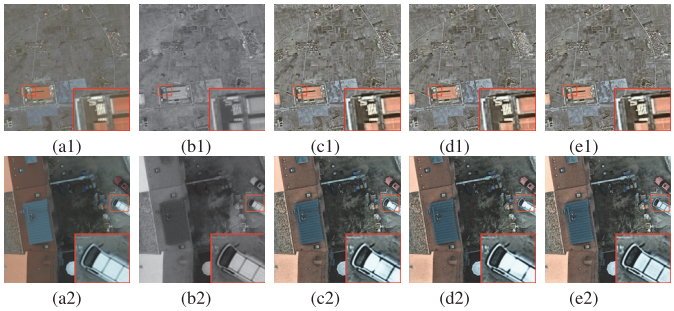


Fig. 12. (a1) and (a2) Visible image. (b1) and (b2) Infrared image. (c1) and (c2) TGFF-w/o-FusionRes. (d1) and (d2) TGFF-SA-TCA. (e1) and (e2) VP-BIL. Visual comparison of TGFF variants on WHU-TA (first group) and Potsdam-TA (second group). Through the collaborative design of cross-attention modules and learnable weighted residual fusion, the TGFF achieves precise semantic alignment while completely preserving visual information. The fused images exhibit sharp textural details and balanced exposure, demonstrating optimal visual fidelity.

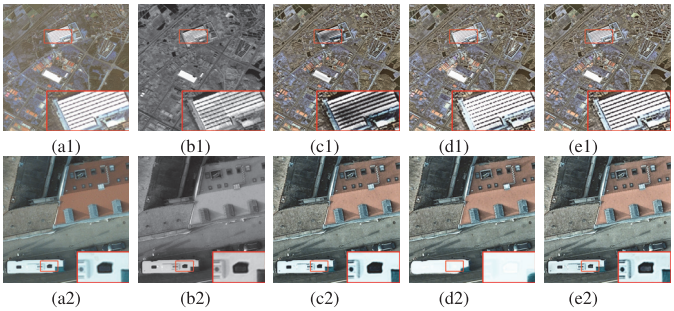


Fig. 13. (a1) and (a2) Visible image. (b1) and (b2) Infrared image. (c1) and (c2) VDTC-w/o-TextRes. (d1) and (d2) VDTC-V2T-Attn. (e1) and (e2) VP-BIL. Visual comparison of VDTC variants on WHU-TA (first group) and Potsdam-TA (second group). The VDTC employs a text residual fusion mechanism integrated with the scale-and-shift mechanism to accurately rectify text description biases, producing fused images devoid of distortion with well-defined object contours and rich hierarchical details, thereby achieving optimal visual quality.

anism between text-guided fusion and vision-aided textual refinement.

On the contrary, the VP-BIL-3iter achieves optimal performance across all quantitative metrics. The corresponding fused images exhibit no discernible visual artifacts such as darkening artifacts or overexposure effects in grid-patterned parcel regions. In particular, textural hierarchies remain well-preserved and distinct, shadow contour boundaries in the desktop regions are clearly delineated with rich fine-grained details, and the overall exposure is well-balanced with high information fidelity and richness. Although VP-BIL-4iter yields marginal improvements over the two-iteration counterpart in several quantitative metrics, its fused images still suffer from notable quality degradations, including global overexposure and saturation-related artifacts in the highlight regions of grid-patterned parcel regions, along with poor clarity in the dark shadow regions of the desktop. Furthermore, computational complexity scales linearly with the iteration number. This phenomenon indicates that excessive iterations introduce redundant feature-level interference. These experimental results confirm that the three-iteration configuration attains an optimal tradeoff between fusion quality performance and computational efficiency, thus identifying it as the optimal iteration configuration for the bidirectional interactive iteration within the TGFF-VDTC.

2) Influence on the Text-Guided Feature Fusion Module: We further perform experiments to verify the effectiveness of the TGFF, for which two variants of the module are designed to conduct comparative evaluations. The quantitative results are summarized in Table V and the qualitative comparisons are provided in Fig. 12. In particular, the TGFF-w/o-FusionRes variant eliminates the residual fusion structure with learnable weights from the original module and instead directly concatenates cross-attention-enhanced features. This modification results in a 0.88% reduction in EN, a 4.40% decrease in AG, and a 1.45% reduction in VIF, respectively. In the corresponding fused images, the discriminability of brick textures on building roofs is significantly diminished, while localized overexposure emerges in the detailed regions of vehicle roofs. This indicates that relying solely on semantically

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TABLE VI

QUANTITATIVE COMPARISON OF VDTC VARIANTS ON WHU-TA AND POTSDAM-TA. THE BEST RESULTS ARE HIGHLIGHTED IN RED AND THE SECOND RESULTS ARE HIGHLIGHTED IN BLUE

Method	WHU-TA						Potsdam-TA					
	EN	SD	SF	AG	SCD	VIF	EN	SD	SF	AG	SCD	VIF
VDTC-w/o-TextRes	6.8967	33.5975	37.3383	13.0502	1.5374	1.0913	7.1152	40.2693	24.3509	8.9246	1.3735	1.5668
VDTC-V2T-Attn	6.8569	33.2260	36.4898	12.6250	1.5636	1.0794	7.0593	40.0883	23.4386	8.5515	1.2827	1.5593
VP-BIL	6.9613	35.6344	39.2144	13.7745	1.6242	1.1057	7.2130	41.9815	25.4023	9.4187	1.3890	1.5875

enhanced features for fusion degrades the original visual information of the source images, thereby compromising the visual authenticity of the fusion results. The TGFF-SA-TCA variant revises the original architecture by adopting dual-modality independent self-attention mechanisms, where text features are concatenated with the self-attention-enhanced features of both modalities and processed via convolution to achieve passive guidance. Notably, this variant yields a 5.54% decline in SF, a 5.21% diminishment in SD, and a 4.07% drop in SCD. In the fused images, the hierarchical representation of brick textures is markedly weakened, with overexposed regions in vehicle areas expanding substantially and fine-grained details nearly obliterated. This outcome highlights that independent self-attention mechanisms are deprived of cross-modal semantic synergy, thereby attenuating the active guiding role of text in visual fusion and consequently introducing fusion bias. Conversely, the fused images generated using the original TGFF architecture exhibit clearly distinguishable brick textures and well-preserved vehicle details, achieving the optimal overall visual performance. Compared to VP-BIL, both variants demonstrate significant degradation in both quantitative performance metrics and visual quality. These findings robustly validate that the residual fusion structure and cross-modal cross-attention mechanism of the TGFF are indispensable for achieving unbiased cross-modal semantic alignment and lossless preservation of source image details.

3) Effectiveness of the Visual-Driven Text Correction Module: To isolate the individual contributions of the text residual correction mechanism and scale-and-shift mechanism to the VDTC module's performance, we construct two variants and conduct verification evaluations. Quantitative results are summarized in Table VI, while qualitative visual results are presented in Fig. 13. VDTC-w/o-TextRes eliminates the text residual fusion structure with learnable weights from the original VDTC module. This variant exhibits a 5.26% reduction in AG, a 4.90% drop in SD, and a 1.53% decline in VIF. Localized dark spot distortions emerge in the corresponding fused images, and the contrast in vehicle sunroof regions is significantly reduced. These results demonstrate that the text residual mechanism is critical for rectifying text description deviations and preserving core semantic information. The absence of the text residual mechanism induces inaccurate text modulation and degrades the module's adaptive robustness across diverse scenarios. VDTC-V2T-Attn replaces the original scale-and-shift mechanism with vision-text cross-attention for textual correction. The fused images generated by this variant present extensive overexposure and whitening artifacts in mesh-patterned texture regions, while fine details in vehicle roof regions are completely obscured due to overexposure. This finding indicates that cross-attention tends to incur redundant attention allocation, highlighting the efficiency superiority of the scale-and-shift mechanism in semantic modulation. On the contrary, the original VDTC generates fused images free from artifacts such as darkening, contrast imbalance, and overexposure. Target contours are sharply defined, with hierarchical fine details well preserved. Both variants achieve significantly inferior performance relative to the original VDTC, thus confirming that the text residual fusion structure and scale-and-shift mechanism of the VDTC are pivotal to mitigating the text description biases.

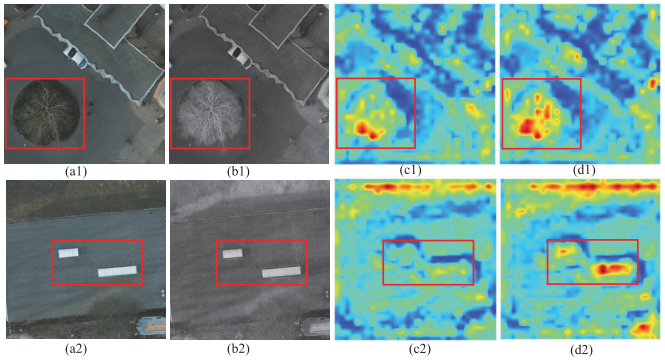


Fig. 14. Visualization of cross-attention heatmaps before and after visual-driven text correction module. (a1) and (a2) Visible image. (b1) and (b2) Infrared image. (c1) and (c2) Heatmap without text correction. (d1) and (d2) Ours.

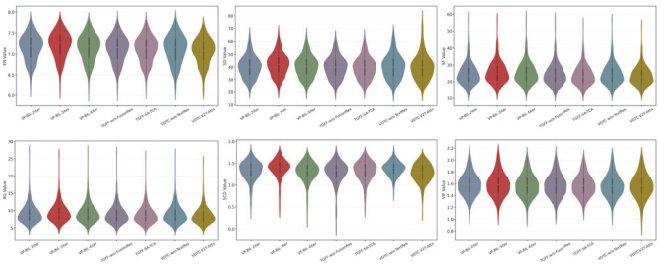


Fig. 15. Violin plots of core metric distributions for various ablation variants on the Potsdam-TA dataset.

To further verify the effectiveness of text-visual alignment at the mechanism level, we visualize the text-visual cross-attention heatmaps before and after the introduction of the VDTC module. The visualization is built upon the cross-attention-based fundamental alignment framework of TGFF, as depicted in Fig. 14. The cross-attention mechanism of TGFF alone can already achieve basic text-visual semantic alignment. Nevertheless, constrained by the coarse-grained nature of text descriptions, certain semantic deviations persist in

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TABLE VII

QUANTITATIVE STATISTICAL RESULTS OF VP-BIL FROM THREE INDEPENDENT REPEATED EXPERIMENTS ON WHU-TA AND POTSDAM-TA

VP-BIL	WHU-TA						Potsdam-TA					
	EN	SD	SF	AG	SCD	VIF	EN	SD	SF	AG	SCD	VIF
Run 1	6.9613	35.6344	39.2144	13.7745	1.6242	1.1057	7.2130	41.9815	25.4023	9.4187	1.3890	1.8575
Run 2	6.9605	35.7190	38.4991	13.4861	1.6385	1.1014	7.1514	41.4007	24.2690	8.9657	1.3952	1.5724
Run 3	6.9381	34.9782	38.6517	13.5341	1.5900	1.1002	7.2081	41.8244	24.7189	9.1116	1.4072	1.5933
Mean	6.9533	35.4439	38.7884	13.5982	1.6176	1.1024	7.1908	41.7355	24.7967	9.1653	1.3971	1.5844
Variance	0.00012	0.10962	0.09462	0.01592	0.00041	0.00001	0.00078	0.06017	0.21709	0.03565	0.00006	0.00008

TABLE VIII

QUANTITATIVE COMPARISON OF DIFFERENT TEMPERATURE COEFFICIENT τ FOR THE VP-BIL MODEL ON WHU-TA AND POTSDAM-TA. THE BEST RESULTS ARE HIGHLIGHTED IN RED AND THE SECOND RESULTS ARE HIGHLIGHTED IN BLUE

Method	WHU-TA						Potsdam-TA					
	EN	SD	SF	AG	SCD	VIF	EN	SD	SF	AG	SCD	VIF
τ=0.4	6.8946	34.0089	37.4103	13.0688	1.5565	1.0876	7.0766	37.3751	24.1293	8.9791	1.0469	1.5654
τ=0.6	6.9613	35.6344	39.2144	13.7745	1.6242	1.1057	7.2130	41.9815	25.4023	9.4187	1.3890	1.5875
τ=0.8	6.8417	33.1957	35.7717	12.5738	1.4792	1.0842	7.0698	37.4229	24.0174	8.9214	1.1091	1.5624

complex remote sensing scenarios with diverse ground objects, manifested as scattered attention distributions and partial misalignment with the actual positions of ground objects such as trees and buildings. On the contrary, with the VDTC module incorporated for fine-grained text semantic rectification, the high-response regions of the attention heatmaps become more concentrated and precisely correspond to the core ground objects in the source images. This phenomenon directly validates that the cross-attention mechanism of TGFF provides fundamental alignment capability, while the VDTC module further rectifies the text semantic deviations on this basis, and their synergy enables more precise text-visual semantic alignment.

4) Statistical Significance Verification: To verify the statistical significance of the metric differences among all model variants in the aforementioned ablation experiments and eliminate the interference of random fluctuations caused by individual experiments or single-sample evaluations, this study visualizes the statistical distributions of six core metrics obtained by all ablation variants on all samples of the Potsdam-TA test set, and constructs the corresponding metric distribution violin plots, as illustrated in Fig. 15. The median is the statistical value that lies in the middle position after sorting the metric values of all test set samples by magnitude, which can reflect the overall performance level of the model on the entire test dataset. A higher median indicates superior comprehensive performance of the model. It can be observed from the visualization results that the proposed model (VP-BIL-3iter) achieves the optimal median across all six core fusion metrics, demonstrating outstanding overall performance. Furthermore, three independent repeated training and evaluation runs are conducted on the complete baseline model VP-BIL to quantify its performance stability, and the quantitative results are summarized in Table VII. The variances of all evaluation metrics remain within a low range, indicating that the inherent stochastic fluctuation of the model is confined to a narrow interval and the proposed model exhibits strong performance stability. These confirm that the performance differences among variants are not caused by random fluctuations, but originate from genuine performance changes brought by alterations to internal module structures or parameter settings, which guarantees the reliability of conclusions drawn from the ablation experiments.

5) Sensitivity Analysis of Hyperparameters: In the proposed VP-BIL model, except for the temperature coefficient τ, which is a manually set hyperparameter, all other weight parameters are learnable parameters that are adaptively optimized by the network during the training process without manual tuning. Therefore, this section only conducts sensitivity analysis on τ, the only manually set key hyperparameter, to verify the rationality and optimality of its value. Comparative tests with different τ values are implemented on the WHU-TA and Potsdam-TA datasets, and the quantitative results are presented in Table VIII. The experimental results reveal that the model achieves optimal performance on all evaluation metrics for both datasets when $\tau = 0.6$, demonstrating the superior fusion performance. An excessively small τ leads to an excessively sharp distribution of cross-modal attention weights, thereby degrading the semantic alignment accuracy. On the contrary, an excessively large τ causes oversmoothing of attention weights and dilutes critical semantic information. Accordingly, this article selects $\tau = 0.6$ as the optimal hyperparameter.

V. CONCLUSION

This article proposes a novel VP-BIL model that eliminates the need for precise textual annotations and effectively mitigates the core issues in IVIF, such as the unidirectional text dependence bias and original visual information loss, thereby establishing a novel paradigm for remote sensing scene fusion. The TGFF model is designed for precise text-visual semantic alignment by leveraging the cross-attention mechanism, while a residual fusion structure with learnable weights jointly ensures semantic consistency and preserves the integrity of

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original details. The VDTC module iteratively calibrates textual biases via deep visual features through multiround iterations, thereby enhancing cross-modal semantic adaptability and achieving text optimization. Furthermore, we construct two remote sensing text–image paired datasets (WHU-TA and Potsdam-TA), providing data support for related research. Experimental results demonstrate that VP-BIL outperforms eight SOTA methods across both datasets on four metrics, accurately restoring critical geographical features. In future work, we will extend it to multisource remote sensing modalities (e.g., SAR and hyperspectral data) and optimize the lightweight interaction mechanism to enhance its real-time applicability.

REFERENCES

[1] J. Ma, Y. Ma, and C. Li, "Infrared and visible image fusion methods and applications: A survey," *Inf. Fusion*, vol. 45, pp. 153–178, Jan. 2019.

[2] W. Zhao, H. Cui, H. Wang, Y. He, and H. Lu, "FreeFusion: Infrared and visible image fusion via cross reconstruction learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 9, pp. 8040–8056, Sep. 2025.

[3] K. Moren and T. Perschke, "Real-time fusion of visible and thermal infrared images in surveillance applications on SoC hardware," in *Counterterrorism, Crime Fighting, Forensics, and Surveillance Technologies II*, vol. 10802. Bellingham, WA, USA: SPIE, 2018, pp. 204–211.

[4] J. Gao, D. Zhang, Q. Wang, Z. Zhao, and X. Li, "Dynamic proxy domain generalizes the crowd localization by better binary segmentation," *Pattern Recognit.*, vol. 172, Apr. 2026, Art. no. 112481.

[5] K. Chen, J. Zhang, C. Liu, Z. Zou, and Z. Shi, "RSRefSeg: Referring remote sensing image segmentation with foundation models," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Aug. 2025, pp. 1070–1074.

[6] T. Zhang et al., "Referring remote sensing image segmentation via bidirectional alignment guided joint prediction," 2025, *arXiv:2502.08486*.

[7] B. Li, H. Dong, D. Zhang, Z. Zhao, H. Sun, and J. Gao, "Exploring efficient open-vocabulary segmentation in the remote sensing," in *Proc. AAAI Conf. Artif. Intell.*, 2026, vol. 40, no. 8, pp. 5982–5991.

[8] J. Gao, D. Zhang, F. Wang, L. Ning, Z. Zhao, and X. Li, "Combining SAM with limited data for change detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5614311.

[9] W. Wang, W. Zhao, H. Wang, and Y. He, "Weakly supervised cross mixer for infrared and visible image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5629413.

[10] H. Xu, J. Ma, J. Jiang, X. Guo, and H. Ling, "U2Fusion: A unified unsupervised image fusion network," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 1, pp. 502–518, Jan. 2022.

[11] F. Zhao, W. Zhao, and H. Lu, "Interactive feature embedding for infrared and visible image fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 9, pp. 12810–12822, Sep. 2024.

[12] W. Tang, F. He, and Y. Liu, "YDTR: Infrared and visible image fusion via Y-shape dynamic transformer," *IEEE Trans. Multimedia*, vol. 25, pp. 5413–5428, 2023.

[13] Z. Wang, Y. Chen, W. Shao, H. Li, and L. Zhang, "SwinFuse: A residual Swin transformer fusion network for infrared and visible images," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–12, 2022.

[14] D. Xu, Y. Wang, X. Zhang, N. Zhang, and S. Yu, "Infrared and visible image fusion using a deep unsupervised framework with perceptual loss," *IEEE Access*, vol. 8, pp. 206445–206458, 2020.

[15] J. Ma et al., "Infrared and visible image fusion via detail preserving adversarial learning," *Inf. Fusion*, vol. 54, pp. 85–98, Feb. 2020.

[16] X. Yi, H. Xu, H. Zhang, L. Tang, and J. Ma, "Text-IF: Leveraging semantic text guidance for degradation-aware and interactive image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27016–27025.

[17] Z. Zhao et al., "Image fusion via vision-language model," 2024, *arXiv:2402.02235*.

[18] C. Cheng et al., "TextFusion: Unveiling the power of textual semantics for controllable image fusion," *Inf. Fusion*, vol. 117, May 2025, Art. no. 102790.

[19] Z. Wang et al., "Multi-text guidance is important: Multi-modality image fusion via large generative vision-language model," *Int. J. Comput. Vis.*, vol. 133, no. 7, pp. 4646–4668, Jul. 2025.

[20] X. Zhang and Y. Demiris, "Visible and infrared image fusion using deep learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 10535–10554, Aug. 2023.

[21] F. Zhao, W. Zhao, L. Yao, and Y. Liu, "Self-supervised feature adaption for infrared and visible image fusion," *Inf. Fusion*, vol. 76, pp. 189–203, Dec. 2021.

[22] F. Zhao, X. Wu, W. Zhao, Z. Zhang, and H. Wang, "High-frequency feature transfer for multispectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5639014.

[23] Z. Zhao et al., "Equivariant multi-modality image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 25912–25921.

[24] Y. Deng, T. Xu, C. Cheng, X.-J. Wu, and J. Kittler, "MMDRFuse: Distilled mini-model with dynamic refresh for multi-modality image fusion," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 7326–7335.

[25] Z. Zhao et al., "CDDFuse: Correlation-driven dual-branch feature decomposition for multi-modality image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 5906–5916.

[26] T. Zhang, J. Zhao, Y. Zhu, G. Cui, and Y. Lyu, "FSATFusion: Frequency-spatial attention transformer for infrared and visible image fusion," 2025, *arXiv:2506.10366*.

[27] J. Liu et al., "DCEvo: Discriminative cross-dimensional evolutionary learning for infrared and visible image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 2226–2235.

[28] L. Jian, X. Yang, Z. Liu, G. Jeon, M. Gao, and D. Chisholm, "SEDRFuse: A symmetric encoder–decoder with residual block network for infrared and visible image fusion," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–15, 2021.

[29] Y. Long, H. Jia, Y. Zhong, Y. Jiang, and Y. Jia, "RXDNFuse: A aggregated residual dense network for infrared and visible image fusion," *Inf. Fusion*, vol. 69, pp. 128–141, May 2021.

[30] R. Hou et al., "VIF-net: An unsupervised framework for infrared and visible image fusion," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 640–651, 2020.

[31] L. Tang, J. Yuan, H. Zhang, X. Jiang, and J. Ma, "PIAFusion: A progressive infrared and visible image fusion network based on illumination aware," *Inf. Fusion*, vols. 83–84, pp. 79–92, Jul. 2022.

[32] H. Bai et al., "Task-driven image fusion with learnable fusion loss," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 7457–7468.

[33] W. Zhao, W. Wang, H. Wang, Y. He, and H. Lu, "CDTFusion: Crossing domain and task for infrared and visible image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 48, no. 2, pp. 1456–1469, Feb. 2026.

[34] J. Liu et al., "Target-aware dual adversarial learning and a multi-scenario multi-modality benchmark to fuse infrared and visible for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 5802–5811.

[35] L. Tang, J. Yuan, and J. Ma, "Image fusion in the loop of high-level vision tasks: A semantic-aware real-time infrared and visible image fusion network," *Inf. Fusion*, vol. 82, pp. 28–42, Jun. 2022.

[36] W. Zhao, S. Xie, F. Zhao, Y. He, and H. Lu, "MetaFusion: Infrared and visible image fusion via meta-feature embedding from object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 13955–13965.

[37] J. Liu et al., "Multi-interactive feature learning and a full-time multi-modality benchmark for image fusion and segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jul. 2023, pp. 8115–8124.

[38] J. Li, D. Li, and S. Savarese, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 19730–19742.

[39] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[40] J. Achiam et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.

[41] M. Zhu, X. Chen, X. Wei, N. Wang, and X. Gao, "TeSG: Textual semantic guidance for infrared and visible image fusion," 2025, *arXiv:2506.16730*.

[42] T. Zhang, J. Zhao, Y. Zhu, and G. Cui, "Coupled degradation modeling and fusion: A VLM-guided degradation-coupled network for degradation-aware infrared and visible image fusion," 2025, *arXiv:2510.11456*.

[43] Z. Wang, J. Zhang, H. Song, M. Ge, J. Wang, and H. Duan, "Highlight what you want: Weakly-supervised instance-level controllable infrared-visible image fusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2025, pp. 12637–12647.

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[44] J. Liu et al., "PromptFusion: Harmonized semantic prompt learning for infrared and visible image fusion," *IEEE/CAA J. Autom. Sinica*, vol. 12, no. 3, pp. 502–515, Mar. 2025.
[45] X. Wang, Z. Hua, and J. Li, "Cross-UNet: Dual-branch infrared and visible image fusion framework based on cross-convolution and attention mechanism," *Vis. Comput.*, vol. 39, no. 10, pp. 4801–4818, Oct. 2023.
[46] Z. Lv et al., "Rethinking cross-modal interaction in multimodal diffusion transformers," 2025, *arXiv:2506.07986*.
[47] A. L. Maas, A. Y. Hannun, and A. Y. Ng, "Rectifier nonlinearities improve neural network acoustic models," in *Proc. ICML*, 2013, vol. 30, no. 1, p. 3.
[48] J. Wehrmann, C. Kolling, and R. C. Barros, "Adaptive cross-modal embeddings for image-text alignment," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 12313–12320.
[49] S. Ma, C. Gong, X. Fan, Y. Ma, and C. Jiang, "RIS-FUSION: Rethinking text-driven infrared and visible image fusion from the perspective of referring image segmentation," in *Proc. ICASSP - IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2026, pp. 11902–11906.
[50] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.
[51] ISPRS. (2016). *2D Semantic Labeling Challenge*. Accessed: Sep. 30, 2021. [Online]. Available: https://www2.isprs.org/commissions/comm3/wg4/semantic-labeling.html
[52] V. Aslantas and E. Bendes, "A new image quality metric for image fusion: The sum of the correlations of differences," *AEU-Int. J. Electron. Commun.*, vol. 69, no. 12, pp. 1890–1896, Dec. 2015.
[53] A. M. Eskicioglu and P. S. Fisher, "Image quality measures and their performance," *IEEE Trans. Commun.*, vol. 43, no. 12, pp. 2959–2965, 1995.
[54] R. Wang, L. Du, Z. Yu, and W. Wan, "Infrared and visible images fusion using compressed sensing based on average gradient," in *Proc. IEEE Int. Conf. Multimedia Expo Workshops (ICMEW)*, Jul. 2013, pp. 1–4.
[55] Y. Han, Y. Cai, Y. Cao, and X. Xu, "A new image fusion performance metric based on visual information fidelity," *Inf. Fusion*, vol. 14, no. 2, pp. 127–135, Apr. 2013.
[56] S. Li, X. Kang, and J. Hu, "Image fusion with guided filtering," *IEEE Trans. Image Process.*, vol. 22, no. 7, pp. 2864–2875, Jul. 2013.
[57] H. Li, T. Xu, X.-J. Wu, J. Lu, and J. Kittler, "LRRNet: A novel representation learning guided fusion network for infrared and visible images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 9, pp. 11040–11052, Sep. 2023.
[58] G. Wu et al., "Every SAM drop counts: Embracing semantic priors for multi-modality image fusion and beyond," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 17882–17891.



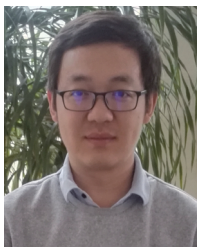
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